Sodium chloride infusion

FORM

LOW CONCENTRATION – Bag containing sodium chloride 0.9%
OR
HIGH CONCENTRATION – Polyfusor containing sodium chloride 2.7%
– Vial containing sodium chloride 30% for dilution

INDICATION

Treatment and prevention of hyponatraemia

DOSE RANGE

AGE	DOSE	ROUTE
Birth – 6 months	2-4 mmol/kg/day* May need higher rates if severely hyponatraemic or if adjusted based on response.	Continuous IV infusion

*Dose ranges listed are typical starting doses and do not represent hard limits. Where higher doses are required in severe hyponatraemia, calculation of sodium deficit can be considered to guide dosing (see calculation below). Caution of the risks of rapid increases in plasma sodium should also be considered.

If serum sodium is >130 mmol supplementation may not be necessary, especially if patient is on PN which can be tailored by pharmacy. Discuss with senior medical staff. Also consider effect of haemolysis on serum sodium results.
Ensure other sources of sodium are taken into account e.g. from PN / oral supplements/ feed

PRESCRIPTION OF CONTINUOUS INFUSION

**** SPECIAL CARE with PRESCRIBING **.**
Flow rates are expressed as ml/**kg**/hour NOT ml/hr

LOW CONCENTRATION

Draw up 50ml of sodium chloride 0.9% from a 500 ml bag into a syringe. (Peripheral)

- Sodium chloride 2 mmol/kg/day = 0.54 ml/**kg**/h
- Sodium chloride 4 mmol/kg/day = 1.08 ml/**kg**/h

HIGH CONCENTRATION – CENTRAL ONLY

THIS PROTOCOL IS RESERVED FOR SEVERE HYPONATRAEMIA OR FLUID RESTRICTED PATIENTS

Draw up 50ml of pre-made sodium chloride 2.7% from a 500 ml bottle into a syringe. (Central)

- Sodium chloride 6 mmol/kg/day = 0.54 ml/**kg**/h
- Sodium chloride 12 mmol/kg/day = 1.08 ml/**kg**/h

Neonatal

RECONSTITUTION Already in solution

DILUTION Sodium chloride 0.9% bag and 2.7% Polyfusor are already diluted for infusion

For sodium chloride 30% concentrate for infusion:

Sodium chloride 30%	3 ml
Sodium chloride 0.9%	Dilute up to 50 ml

This gives a 2.7% sodium chloride solution which can be used as per the above HIGH CONCENTRATION protocol.

COMPATIBILITY

Solution compatibility **Solution incompatibility** **IV Line compatibility** **IV Line incompatibility**	NaCl 0.9% is compatible with PN, and with glucose 5%. For IV line compatibility, NaCl 0.9% is compatible with all drugs where it is listed as a compatible infusion fluid – see individual drug monographs Data on compatibility of NaCl 2.7% is limited. Run through a dedicated line where possible.

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS
Caution when interpreting hyponatraemia in hyperglycaemia

FURTHER INFORMATION
- Ensure plasma sodium does not increase by more than 10mmol/l per day.
- Doses > 6mmol/kg/day may be required but should be guided by serum level and senior medical staff
- Sodium content of 0.9% = 154 mmol/l
- Sodium content of 2.7% = 462 mmol/l
- If plasma sodium levels within range but growth sub-optimal, check urinary sodium (target >40 mmol/l)

LICENSED STATUS Licensed

APPLICABLE POLICIES West of Scotland Neonatal Guidelines

Consult local policy if applicable

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Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Network. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department

Sodium Chloride – worked example using sodium deficit calculation

Baby M
Weight 1.5kg
Plasma Sodium 122mmol/L

Calculating the Sodium Deficit in Extreme Hyponatraemia

Step 1. *Use the equation below to calculate the sodium deficit*

NOTE Plasma sodium should not increase more than 10mmol/L/day therefore if serum Na is 122 then target sodium should NOT exceed 132

Total Sodium deficit (mmol/kg) = (target level – serum sodium level) x 0.6
= (132 – 122) x 0.6
= 6mmol/kg/day

Step 2 *Choose most appropriate route of administration for correction (Peripheral or Central) Consider lines available and fluid status*

Step 3 *Prescribe appropriate dose and volume on infusion chart using information below*

PERIPHERAL ADMINISTRATION (Sodium Chloride 0.9%)

6mmol/kg/day = 1.5ml/kg/hour

= 1.62 x 1.5 (weight)

= 2.43 (2.4ml/hr)

CENTRAL ADMINISTRATION (Sodium Chloride 2.7% Polyfusor)

6mmol/kg/day = 0.5ml/kg/hour (Central)

= 0.54 x 1.5 (weight)

= 0.81ml/hr (0.8ml/hr)